Database Technologies - Assignment 8

Use sub-query to solve following problems.
Note: To solve below queries, use “sales” database

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
mysql> SELECT * FROM orders WHERE cnum = (SELECT cnum FROM customers c WHERE cna
me='Cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.001 sec)
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
mysql> SELECT o.amt, c.cname, c.rating FROM customers c JOIN orders o ON O.cnum
= c.cnum WHERE o.amt > (SELECT AVG(amt) FROM orders);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.001 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
mysql> SELECT o.snum, SUM(o.amt) FROM orders o JOIN salespeople s ON o.snum = s.
snum GROUP BY o.snum, s.sname HAVING SUM(o.amt) > (SELECT MAX(amt) FROM orders);

+------+------------+
| snum | SUM(o.amt) |
+------+------------+
| 1001 |   15382.07 |
+------+------------+
1 row in set (0.001 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
mysql> SELECT cnum, cname, rating FROM customers WHERE rating >= ANY(SELECT rating FROM customers WHERE snum = (SELECT snum from salespeople WHERE sname = 'serres'));
+------+----------+--------+
| cnum | cname    | rating |
+------+----------+--------+
| 2002 | Giovanni |    200 |
| 2003 | Liu      |    200 |
| 2004 | Grass    |    300 |
| 2008 | Cisneros |    300 |
+------+----------+--------+
4 rows in set (0.001 sec)
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
mysql> SELECT * FROM salespeople s WHERE s.city <> ALL(SELECT c.city FROM customers c WHERE c.snum = s.snum);
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.001 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
mysql> SELECT * FROM orders o WHERE o.amt > ANY(SELECT o.amt FROM orders o JOIN customers c ON c.cnum = o.cnum WHERE c.city = 'London');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.002 sec)
```

7. Extract all the orders of Motika.

```
mysql> SELECT * FROM orders o WHERE snum = (SELECT snum FROM salespeople WHERE s
name = 'Motika');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.001 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
mysql> SELECT o.onum, o.amt, o.odate, o.cnum, o.snum FROM orders o JOIN customers c ON c.cn
um = o.cnum WHERE c.cnum = ANY(SELECT c.cnum FROM customers c WHERE city = 'London');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.001 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
ERROR 1054 (42S22): Unknown column 'c.snum' in 'having clause'
mysql> SELECT s.sname, s.snum FROM customers c JOIN salespeople s ON s.snum = c.snum GROUP BY s.snum, s.sname HAVING COUNT
(c.snum) > 1;
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.001 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
mysql> SELECT s.snum, s.sname, s.city,(SELECT COUNT(*) FROM customers c WHERE c.snum = s.sn
um) customer_count FROM salespeople s WHERE s.snum IN (SELECT snum FROM customers GROUP BY
snum HAVING COUNT(*) > 1);
+------+--------+----------+----------------+
| snum | sname  | city     | customer_count |
+------+--------+----------+----------------+
| 1001 | Peel   | London   |              2 |
| 1002 | Serres | San Jose |              2 |
+------+--------+----------+----------------+
2 rows in set (0.002 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
mysql> SELECT * FROM customers WHERE RATING > ANY(SELECT rating FROM customers WHERE city = 'Rome');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.001 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’.

```
mysql> SELECT * FROM orders WHERE amt > ANY(SELECT amt FROM orders WHERE odate = '1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.001 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
mysql> SELECT * FROM orders WHERE amt < ANY(SELECT o.amt FROM orders o JOIN customers c
 ON c.cnum = o.cnum WHERE city = 'san jose');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.001 sec)
```

14. Select those customers whose rating are higher than every customer in London.

```
mysql> SELECT * FROM customers WHERE rating > ANY(SELECT Rating FROM customers WHERE city = 'London');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.001 sec)
```